

W13 Report (Project Part II): Usability Study Report

Team: Guava

Topic: Expedia

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Evaluation and Analysis Results (3~4 pages)

a) Topic:

Our topic is the redesign of the Expedia interface.

b) Evaluation Goals: (0.25 pg)

Will the user be able to use our updated design at least as efficiently as the original design of the interface?

Will the user be able to find certain filters more quickly?

Can the user learn how to use the interface relatively quickly?

c) Summary of evaluation (0.5):

We used observations to evaluate participants' behaviour as they completed the tasks (see below) and used coding sheets to guide data collection. We followed up with an interview. Participants were 6 university students (ages 18-24) who were recruited using convenience sampling.

Tasks

- Find the filter button
- You want it to be Pet-Friendly
- The price per night should be between \$200 - \$500
- It should have at least a 4 star rating
- It should be Fully Refundable
- It should be in Pearl Harbor
- It should have an elevator
- Apply the filters
- Remove Pet-Friendly filter
- Remove all filters

d) Evaluation Rationale (Step 2): (0.25- 0.5pg)

We wanted to determine whether our redesign was more effective than Expedia's original design. Using an observation allowed us to watch the user work through a set of tasks and what their experience is like when interacting with our redesign. We observed their comments, behavior and tone in real time. A semi-structured interview was used to summarize their overall experience, receiving immediate feedback and opinions on how our design compares with the original Expedia design. An interview helped us learn which parts of our design were effective and which required more improvement from the participants' perspective. We also learned why this was the case, understanding more of their usage

patterns with interfaces like Expedia. We planned to do thematic analysis and group and categorize the recurring results into different themes.

We chose university students as they have worked with online platforms for school, possessing at least a basic level of computer literacy (using the Internet). This group is also more likely to have had some past experience using booking websites to book a vacation.

e) Prototyping Rationale (Step 3): (0.25-0.5)

The scope of our med-fi prototype focuses on the redesign of the filtering workflow of Expedia, specifically, on the accommodations page. We found that the current filtering feature is lacking in terms of efficiency and visibility of system status. The main hindrance to the efficiency is the poor layout of the filters. The filters are housed in the side panel of the search results. With little space and a large number of filters, there is a lot of scrolling required when browsing or searching the filters section. To eliminate this problem, in our prototype, we placed the filters in an overlay. The overlay allows the space needed to present the filters in a way that fits in a few scrolls without crowding. This greatly improves the efficiency of locating the filters one wants. As for the lack of visibility of system status, there are two main factors that currently contribute to this problem, with the first being the lack of clarity on what filters are currently applied and second being the absence of communication of what filters are going to be applied. To help the first problem, in our re-design, we created a “Applied Filters” side panel that displays the filters currently applied and also indicates the number of filters applied. The side panel makes this section hard to miss. For the second problem, our solution was to change the filtering workflow so that instead of filters being applied as they are selected, there is an explicit “Apply Filters” button that users select to apply the filters they selected. This not only gives the users more control but it also increases the efficiency as the search results don’t have to be refreshed every time a user adds a filter.

f) Summary of Data, Findings and Analysis (Step 7): (1-1.5 pg)

Qualitative Data

From our data collection and thematic analysis on the data collected, we determined various themes and grouped together the most interesting and significant ones from the observations and follow up interviews using categorization of similar patterns (things users liked, things they found challenging, etc).

The first theme is that most users thought the new interface was very straightforward and simple to use. In the original Expedia interface, there were a lot of different functions going on with different buttons to click. However, on the redesigned mid-fi prototype interface, everything was grouped together in one place on the pop up when clicking the filters button

on the top right or the browse filters button on the left sidebar, making it clear for the user to understand where to go next and what steps they need to take to book their accommodation, with all the filters grouped together in one place instead of all over the page, where it would be hard for them to figure out what to do next.

The second theme is that the filters were very findable, making it more efficient and quicker for the user to finish the task and click on the required filters that they needed. Users liked how all the filters were in one place, clearly labeled and categorized (ex. categorized under amenities, star rating, etc) in the filters window that would pop up when they clicked on the filters/browse filters button. One user felt that they did not need to put in as much effort to find the filters that they needed and thus, resulted in a quicker duration of time to finish the task. Another user also liked how they could just find all the filters at once and apply them beforehand, which made the whole process more efficient and convenient.

The third theme was that most users liked the clearly defined user inputs when selecting the filters and they were easily intuitive to use at first sight except the price range filter. One user stated that they thought the buttons for user inputs of the filters were very easy to use during the task since it was either just clicking a button or using a slider, both very easy to use and easy to learn options when choosing a specific filter. However, another user thought that the sliders were easy to use except for the price range filter. When choosing the price range, users thought that it was hard to get the exact price range required and that it would be easier to use text inputs for the price instead of sliders.

The last theme was that there was only one way to complete the task of booking an accommodation and some users thought that it was constricting and that there was a lack of liberty for them to carry out the steps and choose the filter. Users felt that at times they were forced to click one option or button when in reality, they might have chosen to click something else. For example, when clearing the selected filters, one user intuitively tried to clear the filters and restart by clicking the filters button on the top right. However, they were not allowed to clear the filters that way and had to do it by clicking the clear filters button on the left sidebar and thus, they felt that there could have been easier ways to do the same step (such as when trying to clear the filters).

Quantitative Data

Overall, based on the quantitative data in *Appendix A*, the majority of users had a positive experience based on the averages of the scales of 1-5 of their experiences with this new interface/redesign of Expedia and preferred it compared to the original Expedia interface. Users completed the task of using the filters to book/look for an accommodation very quickly with an average duration of ~ 3 minutes. ***See Appendix A for summary statistics of data.***

g) Conclusions (Step 8): (0.5-1 pg)

Our goal for this usability study report was to pick out a subset of Expedia's existing features to try and improve upon by creating a medium fidelity prototype. We settled on the ability to filter accommodations on Expedia after having identified several layout deficiencies that could be augmented to enhance the overall user experience. With our med-fi prototype, we changed the layout of the filters section by placing them in a modal that users could open and close with ease. Our evaluation questions were centered around exploring whether the user experience on the medium fidelity prototype would be at least as good as the original interface design. To collect data for answering our questions, we conducted observations on participants. They worked through a set of steps for a task example and also provided reflective thoughts afterwards in semi-structured interviews. Some found the process of carrying out the task example on Figma prototypes a bit too constraining in terms of the ordering of steps they could perform. However, our analysis of our qualitative and quantitative data showed that users mostly preferred to work with our prototype over the original interface, as the interface was easy to learn and filters were more findable. Users also liked how clearly visible and defined the user inputs were overall based on our data collection, findings, analysis and the results of the scales in *Appendix A*.

Recommendations and Critique (1 page)

h) Design Recommendations (Step 9): (0.5 pg)

The final interface design is more complete than our lo-fi prototypes. The final prototype emulates Expedia according to the testimony of several participants during the evaluation. Besides the visual and physical realism, we implemented a vertical scope with minimal horizontal scope. Specifically, we *only* developed the filters feature, but implemented several aspects of it. Provided users interacted with the prototype in an appropriate way, the prototype appeared fully functional. However, this means there was only one way to book an accommodation on our interface which users found constrictive. Although the prototype seems complete, each of our input elements need to function completely which would give users flexibility in filtering. This is the vertical scope portion of our design: making the filters function work completely. The purpose of our Expedia redesign was to improve the user experience for specific filtering steps, and users tended to find the filtering feature on our interface more quickly. This indicates that our redesign was effective. Users found it easier to filter for specific criteria. This suggests that our redesign was organized effectively. Furthermore, we may conclude that our prototype is intuitive because none of the participants asked for guidance. During the interviews, users mentioned that they liked the *Apply filters* button too because it made the interface refresh once after every filter was applied, instead of after each user input, improving efficiency and overall user experience.

i) Critique of process (Step 10): (0.5 pg)

We started with two lo-fi prototypes and fleshed out the one we preferred by adding the best features from the other prototype. We did a cognitive walkthrough and iterated on our design based on our findings on where we could improve, making it more functional and complete. We tested our implementation with experimental users to get immediate feedback on our design. This process resulted in a strong med-fi prototype, but upon reflection, there are a few areas for growth. Our original lo-fi prototypes may have been too functional for their intended purpose. The time spent developing the lo-fi prototypes was too much considering the goal was to simply test out very simple ideas. Our lo-fi prototype ultimately required very little change to transform it into a mid-fi prototype. If we could repeat the process, we would spend less time on our lo-fi prototypes and more time on our med-fi prototype. Furthermore, our evaluation process gave us the data we needed to make critical design decisions. We collected data from six users via observation and interview. Users had to complete specific steps using our prototype and tell us their experiences. We recorded the sessions, allowing us to re-watch the videos to find critical incidents. To improve the evaluation process, we should create a script explaining the purpose of the study to streamline the session and help run the evaluations more smoothly.

Appendix A

A.1) Medium Fidelity Prototyping Video Reference

Link to the video: <https://www.youtube.com/watch?v=VurDzqiPTA0>

A.2) Figures and Tables

Total time taken to complete the task	average/mean of ~ 3 minutes
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Figure 1. *Table of the total time taken (in minutes) to complete the task of booking an accommodation*

On a scale of 1 to 5, with one being 1 being extremely difficult and 5 being extremely easy, how hard was it to navigate the interface?	average of ~ 4.3
On a scale of 1 to 5, with one being 1 being extremely difficult and 5 being extremely easy, how hard was it to book an accommodation that matched the criteria?	average of ~ 4.8
On a scale of 1 to 5, with one being 1 being strongly disagree and 5 being strongly agree, do you think your experience with online booking websites helped you learn how to use this interface?	average of ~ 4.8
On a scale of 1 to 5, with one being 1 being strongly disagree and 5 being strongly agree, I had trouble finding at least one filter.	average of ~ 3.0
On a scale of 1 to 5, with one being 1 being strongly disagree and 5 being strongly agree, <i>accomplishing tasks on this interface felt easier than using the previous version of Expedia.</i>	average of ~ 4.2
On a scale of 1 to 5, with one being 1 being strongly disagree and 5 being strongly agree, the feedback from each action matches my expectation.	average of ~ 4.2
On a scale of 1 to 5, with one being 1 being strongly disagree and 5 being strongly agree, I always knew how to do the next step.	average of ~ 4.5

Figure 2. *Table of Interview Questions and their Scale Averages/Mean (On a Scale of 1 - 5)*

A.3) Evaluation Instruments

<https://ln5.sync.com/dl/d0e383080/9nu5d2zj-2spjnhhh-j2vt78t7-9qsf9mfb>

Appendix B - Additional Documentation (separate from Report)

B.1) Adherence to Ethics Protocol form

<https://ln5.sync.com/dl/550cbf780/qr5aznb3-hpa3mpuc-4yq4mkqr-rdewbakw>

B.2) Signed participant consent forms

<https://ln5.sync.com/dl/7d9c9e7e0/5cyzu2qa-qce7cw9y-aseqaipz-s7aep7ga>

B.3) Scanned or digital raw data

<https://ln5.sync.com/dl/d0e383080/9nu5d2zj-2spjnhhh-j2vt78t7-9qsf9mfb>